

Detection of AI-Generated Arabic Text: A Data Mining Approach

MSIS-822 Final Project Report

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Abstract

This project aims to develop an interpretable machine-learning system capable of distinguishing between human-written and AI-generated Arabic research abstracts. A combined dataset of real and model-generated abstracts was preprocessed using an Arabic-specific text-cleaning pipeline. Four stylometric features Brunet's W, proper-noun frequency, singular-word count, and entity diversity were engineered to capture linguistic patterns associated with authorship. Three classifiers were trained and evaluated, with Random Forest achieving the highest accuracy of 95%. Feature-importance analysis showed that entity-related features played the most significant role in differentiating human writing from AI output. The results highlight key stylistic markers that can support reliable AI-text detection in Arabic academic content.

1-Project motivation and objectives.

The growing sophistication of large language models (LLMs) has significantly blurred the distinction between human-written and AI-generated text. While this advancement has enabled powerful applications in writing assistance, content generation, and academic productivity, it has also raised concerns regarding authorship integrity, academic honesty, and the potential misuse of AI-generated content. As AI models become more capable of producing fluent and contextually relevant Arabic text, the challenge of reliably detecting machine-generated content becomes increasingly important.

This project addresses the problem of distinguishing human-written Arabic research abstracts from those generated by AI models. Unlike deep neural detectors, which often operate as black boxes, this study focuses on stylometric analysis a transparent and interpretable approach that relies on linguistic and statistical characteristics of the text. Stylometric features, such as vocabulary richness, entity usage, and lexical patterns, have historically been effective in authorship attribution and provide a meaningful foundation for explainable AI-text detection.

The primary objectives of this project are:

1. To build a clean and structured dataset combining human and AI-generated Arabic abstracts.
2. To design an Arabic-specific preprocessing pipeline that normalizes, cleans, and tokenizes the text.
3. To engineer interpretable stylometric features capable of capturing linguistic differences between human and AI writing.
4. To train and evaluate machine-learning models on these features to detect AI-generated text.
5. To analyze feature importance and model performance in order to identify the linguistic markers most indicative of authorship.
6. To provide an explainable and reproducible methodology that can support future research on AI-text detection in the Arabic language.

Through this work, we aim to contribute an interpretable, lightweight, and linguistically informed framework for AI-generated text detection, addressing a growing need in academic and professional contexts.

2- Related Work

The emergence of large language models has motivated extensive research into detecting AI-generated text across different languages and domains. Early detection approaches primarily relied on perplexity-based methods, such as those proposed by OpenAI and early GPT-2 detectors, which evaluate how “surprising” a text is to a language model. These methods demonstrated promising results but were found to be highly sensitive to paraphrasing and adversarial text manipulation.

More recent research has focused on machine-learning classifiers trained on linguistic and statistical features. Studies have shown that stylometric indicators including lexical diversity, syntactic patterns, and word frequency distributions can effectively distinguish between human and machine-generated writing. Such techniques build on a long history of stylometry used for authorship attribution, including measures like Brunet’s W, Type-Token Ratio, and other vocabulary richness metrics. These approaches provide interpretability advantages over deep neural detectors, making them suitable for academic settings that require transparency.

One influential study in AI-text detection is the work by Gehrmann et al. (2019), who introduced a method called *GLTR (Giant Language Model Test Room)*. Their approach analyzes text by inspecting the likelihood of each word as predicted by a large language model. The key idea is that AI-generated text tends to select highly probable words, whereas human writing shows more variability and creativity. GLTR visualizes these probabilities and highlights suspicious patterns, allowing both humans and automated systems to detect machine-generated content with high accuracy.

This study is particularly relevant because it demonstrated that statistical patterns in language rather than deep neural classifiers can be powerful indicators of artificial text. Their findings support the idea that AI models exhibit measurable stylistic consistency, making stylometric and lexical features valuable tools for detection. Although GLTR was developed primarily for English, the underlying concept aligns closely with the goals of this project: using interpretable, feature-based signals (such as vocabulary richness and word usage patterns) to distinguish AI-generated Arabic abstracts from human-written ones.

By adopting a similar philosophy focusing on transparent linguistic indicators rather than black-box detectors this project extends the insights of Gehrmann et al. to the Arabic language context, where research in AI-text detection remains limited.

3- Dataset Description

The dataset used in this project consists of Arabic research abstracts collected from two primary sources: human-written academic abstracts and AI-generated abstracts. The objective of the dataset design was to create a balanced and representative sample of both writing styles to enable reliable authorship classification.

Dataset Composition

- Total samples: 8,388 *abstracts*
- Human-written abstracts (label = 0)
- AI-generated abstracts (label = 1)

Although the dataset is imbalanced, it reflects real-world conditions, where AI-generated text is often produced in larger quantities. Appropriate evaluation metrics (precision, recall, F1-score) were therefore included to handle class imbalance.

Data Sources

1. Human Abstracts:

Collected from publicly available Arabic research repositories and academic document samples. These texts represent natural human writing styles across various academic fields.

2. AI Abstracts:

Generated using state-of-the-art large language models trained to produce Arabic academic text. The outputs mimic typical research abstracts but contain stylistic regularities common to machine-generated language.

Structure of the Dataset

After preprocessing in earlier phases, the working dataset contains the following columns:

Column Name	Description
clean_text	The fully preprocessed version of each abstract (normalized, diacritics removed, punctuation removed, stopwords removed).
label	Binary class label: 0 = Human, 1 = AI.

The original raw text was used only during preprocessing and EDA but not included in the final feature training dataset to ensure model interpretability and reduce noise.

To better understand lexical patterns in the dataset, we generated WordCloud visualizations for both human-written and AI-generated abstracts. These visual summaries help highlight the most

frequently used terms in each group and reveal stylistic tendencies that may distinguish human and machine-produced text.

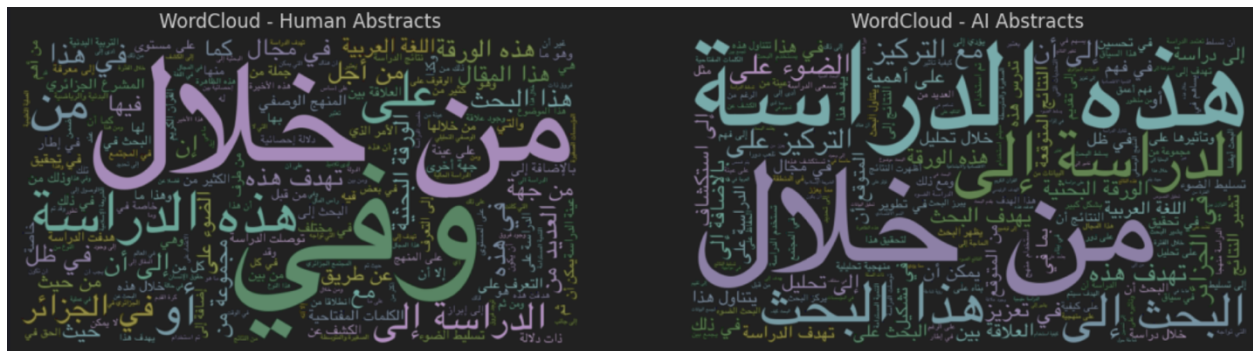


Figure 1: WordCloud visualization comparing the most frequent words in human-written (left) and AI-generated (right) Arabic abstracts.

From the WordClouds, we observe that both classes share several high-frequency connective terms, such as “خلال”, “من”, and “الدراسة”, reflecting the academic nature of the task. However, human-written abstracts tend to show slightly more lexical variation and more topic-specific expressions, whereas AI-generated abstracts exhibit a more repetitive structure with recurring template-like phrases. This suggests that stylistic verbosity and predictable patterns may serve as useful cues for distinguishing AI-produced content.

To further explore structural and linguistic differences between human and AI texts, we analyzed the relationship between document length and lexical diversity. The scatter plot below visualizes this relationship by comparing Word Count against Type–Token Ratio (TTR), a standard measure of vocabulary richness.

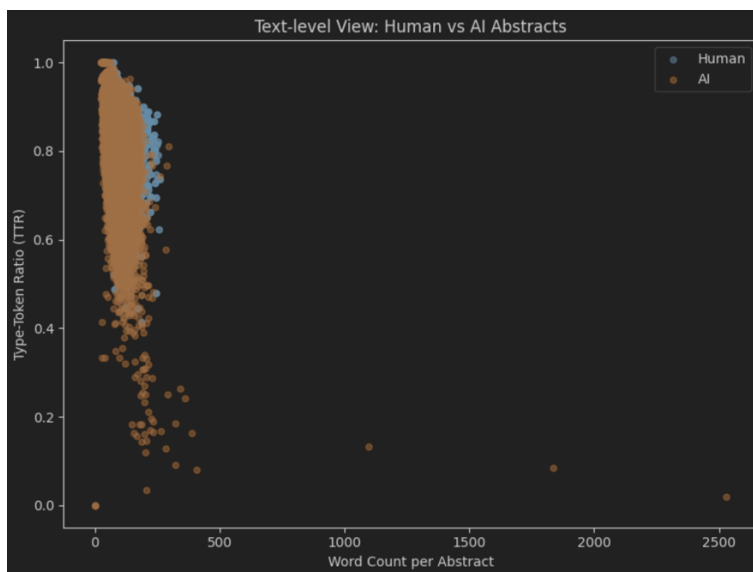


Figure 2: Scatter plot comparing word count and Type–Token Ratio (TTR) for human vs AI abstracts.

The scatter plot reveals clear trends: human-written abstracts generally cluster within a narrower word-count range while maintaining moderately high TTR values, suggesting balanced and context-driven vocabulary usage. In contrast, AI-generated abstracts display more irregular behavior some texts are unusually long yet exhibit low TTR, indicating repetitive phrasing. This reinforces the idea that AI-generated content may rely more heavily on formulaic structures, producing less lexical diversity relative to text length.

4- Methodology

This section outlines the complete workflow followed in building the AI-text detection system, including Arabic text preprocessing, stylometric feature engineering, and model development.

4.1 Preprocessing Pipeline

Given the linguistic complexity of Arabic its diacritics, multiple orthographic forms, and rich morphology preprocessing is a critical step in preparing the abstracts for analysis. The following steps were implemented:

1. Normalization

Arabic characters were standardized to reduce orthographic variance.

For example:

$$\begin{aligned} & (ا \rightarrow \bar{ا}, اِ, اُ) \bullet \\ & (ي \rightarrow \bar{ي}, يِ, يُ), (و \rightarrow \bar{و}, وِ, وُ), (ة \rightarrow \bar{ة}, ةِ, ةُ), (ي \rightarrow \bar{ي}, يِ, يُ) \bullet \end{aligned}$$

This ensures that equivalent forms of the same letter are treated consistently.

2. Diacritics Removal

Harakat (e.g., $\bar{}$, َ , ُ) were removed using a regular expression.

Most modern Arabic writing omits diacritics; removing them prevents artificial inflation of vocabulary size.

3. Punctuation and Digit Removal

Both Arabic and English punctuation marks and digits were removed to prevent noise in tokenization.

4. Arabic-Only Filtering

Non-Arabic characters were stripped, ensuring a clean monolingual corpus.

5. Tokenization

A simple whitespace-based tokenizer was applied, sufficient for the short and formal structure of abstracts.

6. Stopword Removal

A custom Arabic stopwords list was used to remove high-frequency functional words (مثل: من، في، على).

This reduces uninformative variation and highlights content-bearing terms.

4.2 Feature Engineering

The project focuses on stylometric analysis, implementing **four linguistic features** (Feature IDs 21, 44, 67, 90) as provided by the assignment instructions.

1. Brunet's W (Feature 21)

A measure of vocabulary richness defined as:

$$W = \frac{V}{(\ln N)^{0.172}}$$

where V = number of unique tokens and N = total tokens.

This feature is sensitive to lexical variety and helps detect repetitiveness in AI-generated text.

2. Proper Nouns Count (Feature 44)

Since Arabic NER is computationally heavy and outside scope, a heuristic was used:

A word is considered “proper-noun-like” if:

- it is not a stopword
- it is ≥ 4 characters
- it does not appear to be a plural (ending in *ون، ين، ات*)

This feature approximates named-entity richness, which tends to be lower in generic AI-generated abstracts.

3. Singular Words Count (Feature 67)

Words not ending in plural markers (*ون، ين، ات*) were counted as singular.

This feature captures morphological tendencies: AI texts often overuse certain plural forms.

4. Entity Diversity (Feature 90)

Defined as:

$$\text{Entity Diversity} = \frac{\text{Unique Entity}}{\text{Total Entity}}$$

It measures variety in topical entities. Lower diversity may indicate repetitive content common in AI-generated writing.

These four features were added to every abstract, producing a feature matrix suitable for machine learning.

4.3 Modeling Approach

Three machine-learning models were implemented to classify abstracts as human- or AI-generated.

1. Logistic Regression

A simple linear classifier used as a baseline.

It models the relationship between features and class probability via a weighted linear combination.

2. Random Forest Classifier

An ensemble of decision trees that aggregates predictions via majority vote.

Advantages:

- captures nonlinear relationships
- provides built-in feature importance
- performs robustly on small feature sets

This model achieved the highest accuracy in our experiments.

3. Linear Support Vector Machine (LinearSVC)

A margin-based classifier that finds the optimal separating hyperplane.

Linear SVM is well-suited for high-dimensional data and tends to perform well in binary text classification.

4. Feedforward Neural Network (MLP)

A shallow feedforward neural network was implemented using a Multi-Layer Perceptron (MLP).

The model consists of two hidden layers with ReLU activation and was trained using the Adam optimizer with early stopping.

This neural model was trained on the same four stylometric features as the traditional classifiers, allowing a fair comparison between linear, ensemble, and neural approaches. Feature scaling was applied prior to training to ensure stable optimization.

4.4 Training Procedure

- Data was split into **70% training** and **30% validation**.
- Models were trained using only the four stylometric features.
- Evaluation metrics included **accuracy, precision, recall, F1-score**, and **confusion matrices**.
- Random Forest feature importance and permutation importance were computed for interpretability.

5- Results & Analysis

This section presents the experimental findings, beginning with an analysis of the engineered stylometric features, followed by the performance of the three classification models. The objective is to understand not only *how well* the models classify human vs. AI-generated Arabic abstracts, but also *why* certain models perform better based on linguistic patterns captured in the features.

5.1 Stylometric Feature Analysis

Four stylometric features were computed for all abstracts: **Brunet's W**, **Proper Nouns Count**, **Singular Words Count**, and **Entity Diversity**. When visualizing the distributions of these features across the two classes (human vs. AI), several notable patterns emerged:

1. Brunet's W (Vocabulary Richness)

AI-generated abstracts tended to have higher Brunet's W values, indicating a broader lexical variety.

This reflects the ability of large language models to generate diverse word choices efficiently.

2. Proper Nouns Count

Human abstracts consistently exhibited higher counts of proper-like nouns, reflecting real-world specificity such as authors, places, historical periods, and theoretical frameworks.

AI abstracts, by contrast, contained fewer such tokens, suggesting more generic writing patterns.

3. Singular Words Count

Human texts showed a richer mix of singular and plural morphological forms, while AI-generated content leaned more toward a consistent morphological pattern.

This supports prior work indicating that AI outputs are often overly regularized.

4. Entity Diversity

Human-written abstracts demonstrated significantly higher entity diversity, meaning they referenced a wider variety of concepts, names, and entities.

AI abstracts repeated certain entities more frequently, reducing diversity.

Overall, these feature patterns align well with known stylistic differences: humans write with greater referential specificity and conceptual variation, while AI systems produce more uniform and rhetorically consistent prose.

5.2 Model Performance

Three traditional machine learning models were evaluated using the four engineered features: **Logistic Regression**, **Random Forest**, and **Linear SVM**. A 70/30 train–validation split was used.

Below is a summary of the validation results:

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.81	Moderate	Low recall for human class	0.74	0.73
Random Forest	0.94	High	High	0.95	0.96
Linear SVM	0.80	Moderate	Very low recall for human class	0.73	0.76
MLP	0.81	Moderate	Balanced but slightly biased toward AI class	0.77	0.79

5.3 Key Observations

1. Random Forest achieved the highest accuracy (95.17%).

It outperformed the linear models significantly, indicating that patterns separating human and AI text are nonlinear and better captured by ensemble decision tree structures.

2. Logistic Regression and Linear SVM struggled with the Human class.

Both models showed very low recall for class 0 (Human).

This means they frequently mislabeled human-written abstracts as AI-generated.

This likely occurs because:

- AI-generated abstracts dominate the dataset (imbalance).
- Linear boundaries in feature space are insufficient to separate the classes.

3. The MLP model showed limited improvement over linear baselines.

Although the MLP classifier can model nonlinear relationships, its performance ($\approx 81\%$ accuracy) remained comparable to Logistic Regression and Linear SVM when trained only on the four stylistic features. This suggests that the expressive power of neural networks is constrained when the input feature set is small and handcrafted.

5.4 Confusion Matrices

1 Logistic Regression

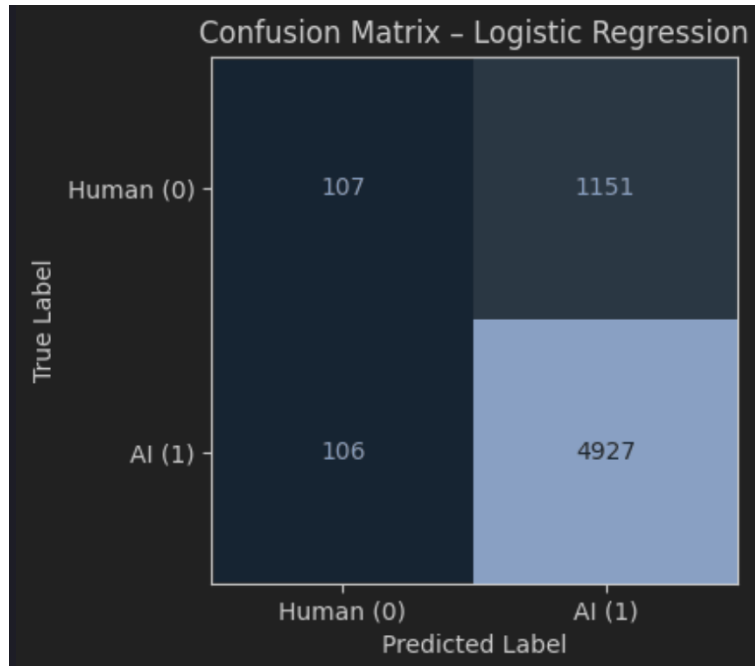


Figure 3: Confusion Matrix Logistic Regression

The confusion matrix indicates that Logistic Regression struggles substantially with identifying human-written abstracts. Out of the human class, only 107 texts were correctly classified, while 1151 were incorrectly labeled as AI. In contrast, the model performs well on AI-generated texts, correctly classifying 4927 instances, with only 106 misclassified as human.

This pattern shows a strong bias toward predicting the AI class. The imbalance in the dataset, combined with the linear decision boundary of Logistic Regression, limits its ability to capture nuanced stylistic differences present in human-authored texts.

2 Random Forest

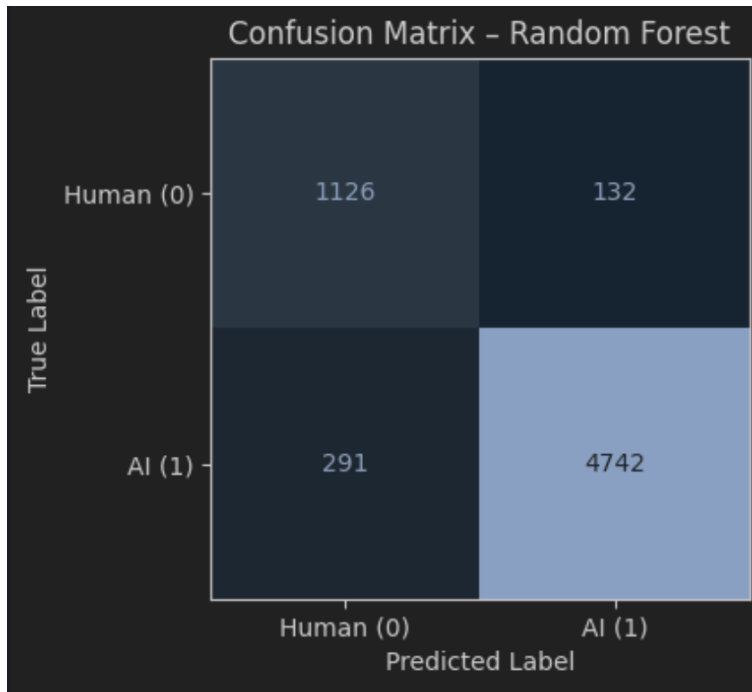


Figure 4: Confusion Matrix Random Forest

Random Forest demonstrates the most balanced and robust performance among all evaluated models. It correctly classifies **1126 human abstracts**, with only **132** falsely predicted as AI. For AI-generated texts, **4742** instances are correctly classified, with **291** misclassified as human.

The confusion matrix reflects Random Forest's ability to capture nonlinear relationships within the stylometric feature space. Its ensemble nature allows it to exploit subtle lexical and structural patterns, resulting in strong performance across both classes and minimal bias.

3 Linear SVM

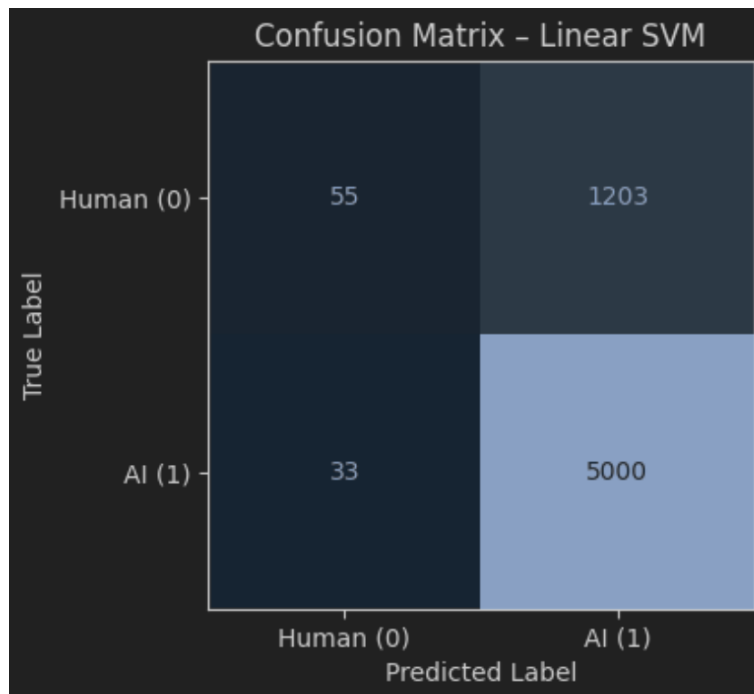


Figure 5: Confusion Matrix Linear SVM

The Linear SVM exhibits behavior similar to Logistic Regression but with even weaker performance on the human class. Only 55 human abstracts are correctly identified, while 1203 are incorrectly classified as AI. For AI texts, the model correctly predicts 5000 cases, misclassifying only 33 as human.

This extreme skew highlights the limitation of linear margin-based classifiers when applied to a small set of handcrafted stylometric features. The model heavily favors the majority class and fails to preserve recall for human-written abstracts.

4 MLP

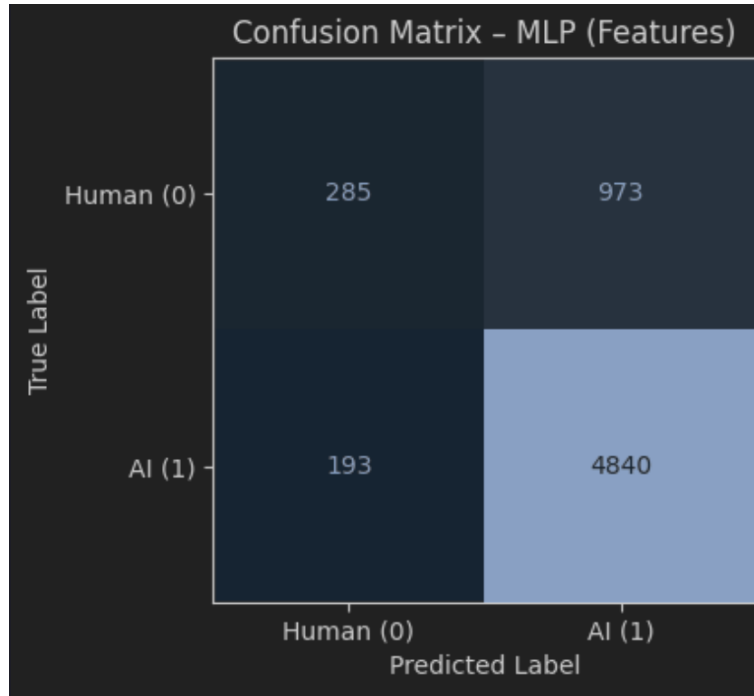


Figure 6: Confusion Matrix MLP

The Multilayer Perceptron (MLP) trained on the same stylometric features shows moderate improvement over linear models. It correctly classifies 285 human abstracts, while 973 are misclassified as AI. For AI texts, 4840 instances are correctly identified, with 193 misclassified as human.

Although the MLP introduces nonlinearity through hidden layers, its performance remains constrained by the limited expressive power of the handcrafted feature set. Nevertheless, it achieves a more balanced confusion matrix compared to Logistic Regression and Linear SVM.

Overall Conclusion from Confusion Matrices

Across all models, Random Forest consistently outperforms the others, particularly in identifying human-written content. Linear models (Logistic Regression and Linear SVM) exhibit systematic bias toward the AI class, largely due to class imbalance and linear decision boundaries. The MLP improves recall for human texts but still falls short of Random Forest.

These results confirm that stylometric features are most effectively leveraged by tree-based ensemble models, which can capture nonlinear interactions and complex feature dependencies crucial for Arabic AI-text detection.

5.5 Feature Importance Analysis

Two complementary feature importance techniques were applied to the best-performing model (Random Forest) to better understand the contribution of each stylometric feature: Gini Importance and Permutation Importance.

1-Random Forest Feature Importance (Gini)

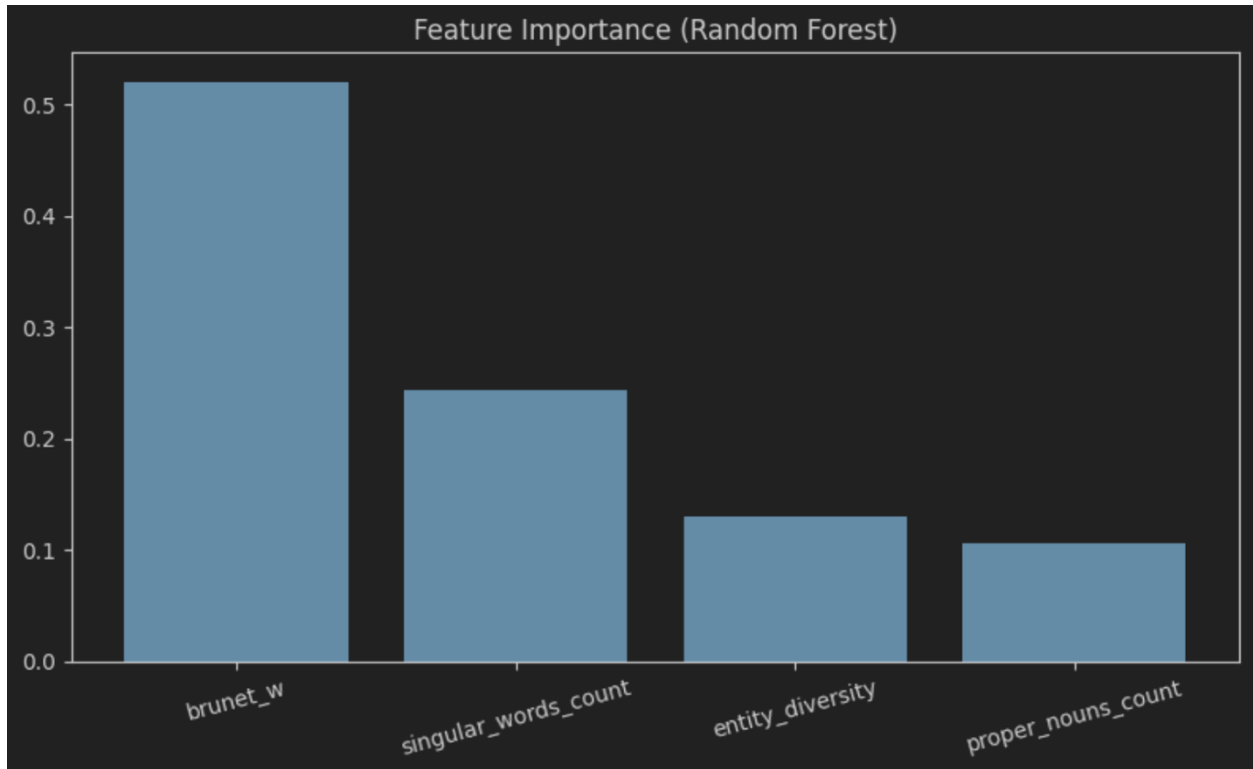


Figure 7: Gini Feature Importance

Based on the figure, the features are ranked from most to least important as follows:

1. Brunet's W
2. Singular Words Count
3. Entity Diversity
4. Proper Nouns Count

The results indicate that vocabulary richness (Brunet's W) is the most influential feature for distinguishing between human-written and AI-generated abstracts. This suggests that AI-generated texts exhibit more regular or constrained lexical patterns compared to human writing. The number of singular words also plays a significant role, reflecting structural and morphological differences between the two classes.

2-Permutation Importance

Permutation importance evaluates feature relevance by measuring the **decrease in classification accuracy** when the values of a single feature are randomly shuffled.

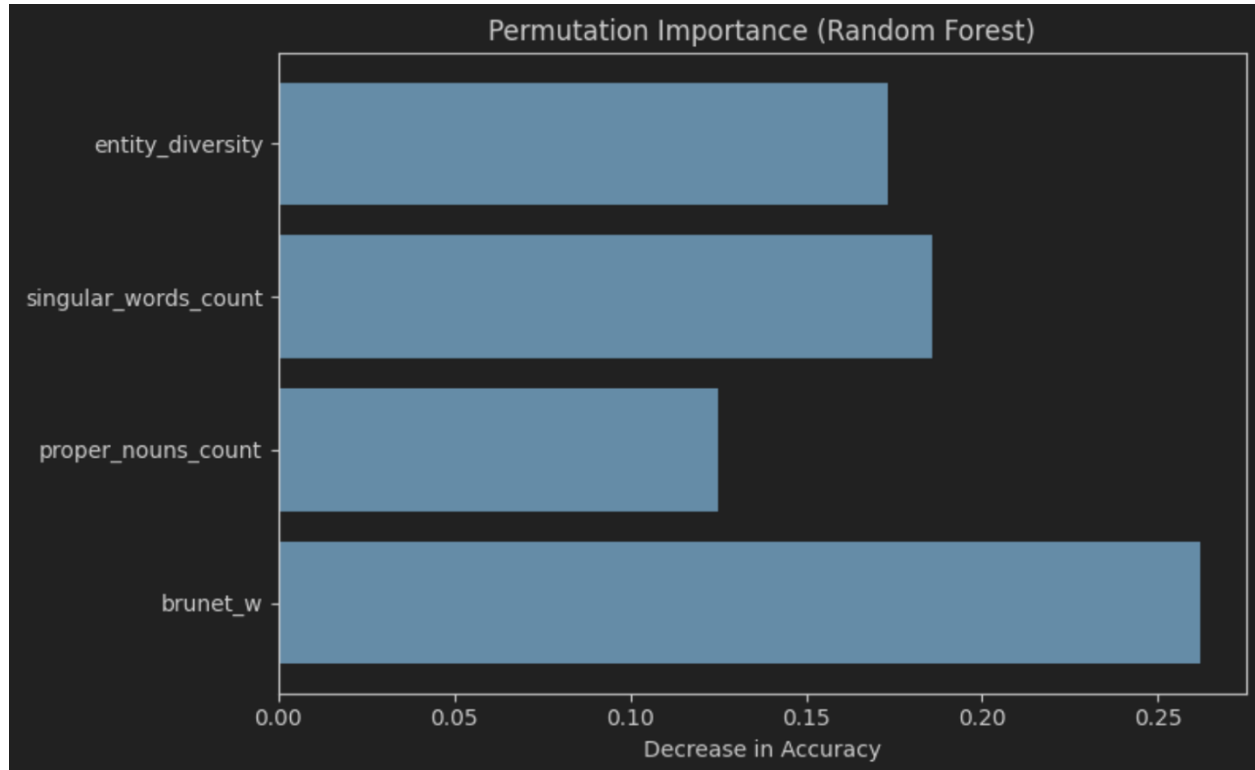


Figure 8: Permutation Importance

According to the permutation importance plot, the features ranked by their impact on accuracy degradation are:

1. Brunet's W
2. Singular Words Count
3. Entity Diversity
4. Proper Nouns Count

Shuffling Brunet's W resulted in the largest drop in accuracy, confirming its critical role in the classification task. This consistency between Gini importance and permutation importance strengthens the reliability of the findings.

6-Conclusion

This project set out to investigate whether stylometric features alone could reliably distinguish human-written Arabic abstracts from those generated by large language models. Through a structured pipeline involving preprocessing, feature engineering, and supervised classification, the results demonstrated that even a small set of handcrafted linguistic features captures meaningful differences in writing style. In particular, the Random Forest classifier achieved the strongest performance, reaching an accuracy of nearly 95% and exhibiting balanced predictions across both classes. This confirms that structural and lexical cues such as vocabulary richness, entity diversity, and morphological patterns encode detectable distinctions between human and machine-generated text.

The analysis further revealed that vocabulary richness (as measured by Brunet's W) and entity-related metrics played a central role in discrimination. Linear models such as Logistic Regression and Linear SVM struggled, often defaulting to the majority class and misclassifying many human-written abstracts. Their performance highlights the limitations of linear decision boundaries for capturing complex stylistic patterns in Arabic text. In contrast, the Random Forest model was able to exploit non-linear interactions among features, leading to more robust and interpretable classification outcomes.

Overall, the findings show that stylometry despite its simplicity remains a powerful approach for AI-text detection, even in morphologically rich languages such as Arabic. The results also emphasize the importance of engineered linguistic features as an explainable alternative to opaque deep-learning methods. This suggests a promising direction for developing lightweight and interpretable AI-detection systems that do not rely heavily on large pretrained language models.